



FACULTY OF ACCOUNTING AND INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

INTEGRATIVE PROGRAMMING AND TECHNOLOGIES III

TEST 3 Memo

DATE: 23 May 2024

TIME: 40 minutes

MARKS: 30

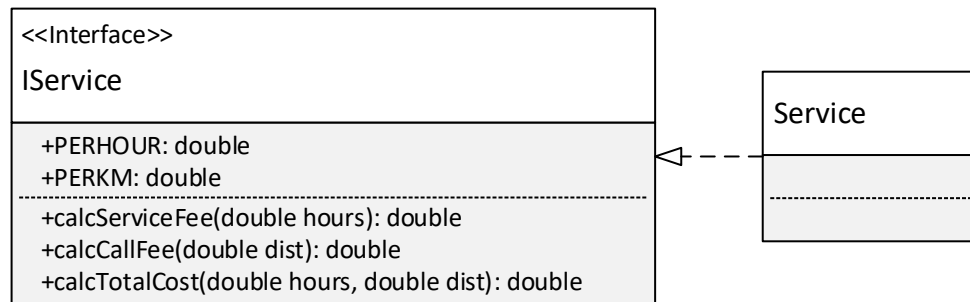
EXAMINER: BOAMAH-ABU, CHARLES (PhD)

NO. OF PAGES: 4 (Including cover page)

INSTRUCTIONS/REQUIREMENTS

1. This examination is closed book
2. Answer all questions
3. Where stated "briefly", please give a short answer
4. Ensure your handwriting is readable

NLM Locksmith offers call-out service to clients. You are to write an RMI application which calculates the total fee for a call-out service. The UML diagram below shows the remote interface and remote class of the application.



The interface contains:

- calcServiceFee(double hours): takes the number of **hours** worked as parameter and returns the service fee at R200 per hour
- calcCallFee(double dist): takes the **distance** to the client as parameter and returns the call-out fee at R50 per kilometre
- calcTotal(double hours, double dist): takes the **hours** and **distance** as parameters, calls the appropriate methods, and returns the total fee (i.e., the service fee and call out fee)
- PERHOUR is a constant (the rate per hour, i.e., R200)
- PERKM is a constant (the rate per kilometre, i.e., R50)

Use the information above to answer **Questions 1 to 5** below.

1. Write the code for the remote IService.

(11 marks)

Answer:

```

public interface IService extends Remote {
    public final double PERHOUR=200;
    public final double PERKM=50;
    public abstract double calcServiceFee(double hours) throws Exception;
    public abstract double calcCallFee(double dist) throws Exception;
    public abstract double calcTotal(double hours, double dist) throws Exception;
}
  
```

2. Write the class header for Service.

(3 marks)

3. Write the code for the following methods in the class Service.

- a. Default constructor.

(2 marks)

- b. calcTotal(double hours, double dist).

(4 marks)

Answer

```

public class Service extends UnicastRemoteObject implements IService {
    public Service() throws Exception {
    }
    public double calcServiceFee(double hours) throws Exception {
        return hours * PERHOUR;
    }
    public double calcCallFee(double dist) throws Exception {
    }
}
  
```

```

        return dist*PERKM;
    }
    public double ✓calcTotal(double hours, double dist) throws Exception ✓{
        return calcServiceFee(hours)+calcCallFee(dist); ✓✓
    }
}

```

4. Complete the code segment below for a class **Server** which creates, registers and binds an instance of **Service** (called **s**) with the reference “**service**”. Assume the class **Service** has been implemented; use port **2024**. (5 marks)

```

...
...
public class Server {
    public static void main(String[] args) throws Exception{
        _____
        _____
        _____
    }
}

```

Answer

```

...
...
public class Server {
    public static void main(String[] args) throws Exception{
        ✓Service s=new Service();
        Registry r=LocateRegistry.createRegistry(2024); ✓✓
        r.rebind("service", s); ✓✓
    }
}

```

5. Complete the code segment below for a class **Client** which creates an instance (called **is**) of the remote object, calls a method of **is** (with hours=2hours and distance=2km) and displays the total fee. Assume **IService** has been implemented, the server is “**localhost**” and the port is **2024**. (5 marks)

```

...
...
public class Client {
    public static void main(String[] args) throws Exception{
        _____is=_____
        double hours=2;
        double dist=2;
        System.out.println("Hours="+hours);
        System.out.println("Distance(km)="+dist);
        System.out.println("Total fee=R"+_____);
    }
}

```

Answer:

```
...
...
public class Client {
    public static void main(String[] args) throws Exception{
        ✓IService is=(IService)Naming.lookup("rmi://localhost:2024/service");✓✓
        double hours=2;
        double dist=2;
        System.out.println("Hours="+hours);
        System.out.println("Distance(km)="+dist);
        System.out.println("Total fee=R"+is.calcTotal(2, 2));✓✓
    }
}
```